

Appendix

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A Dataset Details

Table 1 summarises the six benchmark datasets used in this study, and Table 2 provides representative sample queries from each. All datasets were sampled to a maximum of 50 queries each, yielding 291 total query–context–answer triples after filtering records with empty context fields. Query-type labels were assigned automatically: HotpotQA and SQuAD v2 used meta-data flags (`type` and `is_impossible`) embedded in the dataset; all remaining datasets contain only single-hop questions as defined by their original design, confirmed by manual inspection of a random 20-item subset.

Table 1: Benchmark datasets included in the balanced evaluation corpus.

Dataset	Domain	Query Types	Source (HF path)
HotpotQA	General	Multi-hop, Comparative	hotpot_qa/distractor
SQuAD v2	General	Single-hop, Unanswerable	rajpurkar/SQuAD-v2.0
PubMedQA (labeled)	Medical	Single-hop	qiaojin/PubMedQA/pqa_labeled
PubMedQA (artif.)	Medical	Single-hop	qiaojin/PubMedQA/pqa_artific
FinQA	Finance	Single-hop	ibm/finqa
FinanceBench	Finance	Single-hop	PatronusAI/financebench

*Nine records excluded due to empty context after preprocessing.

B Chunking Strategy Specifications

Six chunking strategies were evaluated, combining two methods (fixed-size and semantic) with three token-budget sizes (128, 256, 512). Table 3 provides full specifications. All token counts use the `cl100k_base` tiktoken encoding (GPT-3/4 vocabulary). Fixed strategies produce non-overlapping windows; semantic strategies respect paragraph and sentence boundaries before falling back to fixed chunking for units that exceed the token budget.

C LLM-as-Judge Prompt

The following prompt was used verbatim for all LLM-as-judge evaluations. Scores were elicited at temperature 0 using `openai/gpt-4o-mini` via the

OpenRouter API. Raw integer scores (1–5) were normalized to $[0, 1]$ by the transformation $s = (r - 1)/4$.

You are an expert evaluator assessing the correctness of a RAG system’s answer.

Task

Rate how correctly the RESPONSE answers the QUESTION, compared to the REFERENCE ANSWER.

Inputs

- QUESTION: {question}
- REFERENCE ANSWER: {ground_truth}
- RESPONSE: {response}

Scoring Rubric (1-5)

5 = The response contains the correct answer with the same essential information as the reference. Minor wording differences or additional correct elaboration are acceptable.

4 = The response is correct but misses minor details from the reference, or adds minor inaccuracies alongside the correct answer.

3 = Partially correct. Captures some key information but misses important details or includes significant inaccuracies.

2 = Mostly incorrect. Attempts to answer but misses the main point or contradicts the reference on key facts.

1 = Completely incorrect, does not answer the question, or contradicts the reference answer entirely.

Important

- Focus on SEMANTIC correctness, not lexical overlap.
- A verbose response that contains the correct answer should score 5.
- Short reference answers (e.g. “yes”, “no”, a name, a date): the response must contain or clearly convey that answer.
- If the question is unanswerable and the reference indicates this, a response that appropriately declines should score highly.

Respond with ONLY valid JSON: {"score": <int 1-5>,
"reasoning": "<one sentence>"}

D Query-Type Classification Prompt

For datasets without built-in query-type metadata, the following prompt was used to classify each question via `openai/gpt-4o-mini` at temperature 0.

Classify the following question into exactly one of these query types:
single-hop, multi-hop, comparative, unanswerable.
A single-hop question requires one fact.
A multi-hop question requires combining facts from multiple sources.
A comparative question requires comparing two or more entities.
An unanswerable question cannot be answered from typical reference material.
Respond with ONLY the query type label.

Question: {query}

E RAGAS Metric Definitions

Four retrieval and generation metrics were computed using the RAGAS framework [1] alongside the LLM-as-judge score. All RAGAS metrics are bounded $[0, 1]$ (higher = better).

Faithfulness Proportion of claims in the generated answer that are directly supported by the retrieved context. Computed by decomposing the answer into atomic claims and checking each against the context with an LLM verifier.

Answer Relevancy Cosine similarity between the embedding of the question and embeddings of synthetic questions reverse-engineered from the answer. Measures whether the answer actually addresses the question asked.

Context Precision Mean reciprocal rank of relevant context chunks among the top- k retrieved chunks, weighted by position. Measures how highly the retriever ranks the most relevant passages.

Context Recall Fraction of ground-truth answer claims that can be attributed to the retrieved context. Measures retrieval completeness.

F Full ANOVA Results

G Tukey HSD Post-Hoc Tables

H Targeted Contrast: Finance vs. General Domain

A Welch’s t -test was conducted on the difference score (LLM Judge – Faithfulness) to isolate retrieval-generation divergence between the Finance and General domains. Results are reported in Table 8.

This result indicates that in the Finance domain the LLM judge consistently awarded lower scores relative to faithfulness compared with the General domain, suggesting that Finance queries impose a correctness penalty beyond what retrieval quality alone explains—likely reflecting the numerical and domain-specific vocabulary demands of financial QA.

I RAG Pipeline Overview

The end-to-end RAG evaluation pipeline proceeded as follows:

1. **Document ingestion.** Each query’s gold context passage was chunked according to one of the six strategies (Appendix B).
2. **Embedding and indexing.** Chunks were embedded with `text-embedding-3-small` (OpenAI, $d = 1536$) and stored in a Qdrant vector collection, one collection per strategy.
3. **Retrieval.** At inference time, the query was embedded with the same model and the top- $k = 5$ nearest chunks were retrieved via cosine similarity.

4. **Generation.** Five LLMs (GPT-4o-mini, Claude 3.5 Sonnet, Gemini 2.5 Pro, Llama 3.1 70B, Mistral Large) generated answers from a fixed prompt template: “*Based on the context below, answer this question: {query}\n\nContext:\n{context}*”.
5. **Evaluation.** Each (question, context, answer) triple was scored by the LLM-as-judge (Appendix C) and by the four RAGAS metrics (Appendix E), both using `openai/gpt-4o-mini` via OpenRouter at temperature 0.

All inference and evaluation calls were routed through the OpenRouter API. LLM responses were generated with default temperature settings per model.

References

- [1] Es Sadig, S., et al. (2023). *RAGAS: Automated Evaluation of Retrieval Augmented Generation*. arXiv:2309.15217.

Table 2: Representative sample queries drawn from the six benchmark datasets. Each row shows one question–answer pair as presented to the RAG pipeline. “N/A” denotes unanswerable items for which no reference answer exists. Medical answers are truncated for brevity; full long-form answers were used during evaluation.

Dataset	Domain	Query Type	Question	Reference Answer
HotpotQA	General	Comparative	Were Scott Derrickson and Ed Wood of the same nationality?	Yes
HotpotQA	General	Multi-hop	What government position was held by the woman who portrayed Corliss Archer in the film <i>Kiss and Tell</i> ?	Chief of Protocol
SQuAD v2	General	Single-hop	In what country is Normandy located?	France
SQuAD v2	General	Unanswerable	Who gave their name to Normandy in the 1000s and 1100s?	N/A
PubMedQA	Medical	Single-hop	Do mitochondria play a role in remodelling lace plant leaves during programmed cell death?	Yes — mitochondrial dynamics were observed to progress as PCD advanced . . .
PubMedQA (artif.)	Medical	Single-hop	Are group 2 innate lymphoid cells (ILC2s) increased in chronic rhinosinusitis with nasal polyps or eosinophilia?	Yes — ILC2s are elevated in CRSwNP and may drive nasal polyp formation . . .
FinQA	Finance	Single-hop	What is the average payment volume per transaction for American Express?	127.40
FinanceBench	Finance	Single-hop	What is the FY2018 capital expenditure amount (in USD millions) for 3M?	\$1,577.00

Table 3: Chunking strategy specifications. “Max tokens” denotes the upper token limit per chunk; fixed strategies use exact-size windows with no overlap.

Strategy	Method	Max tokens	Overlap	Boundary logic
fixed_128	Fixed	128	0	Hard token window
fixed_256	Fixed	256	0	Hard token window
fixed_512	Fixed	512	0	Hard token window
semantic_128	Semantic	128	—	Paragraph → sentence → fixed
semantic_256	Semantic	256	—	Paragraph → sentence → fixed
semantic_512	Semantic	512	—	Paragraph → sentence → fixed

Table 4: Type III ANOVA Results for LLM-as-Judge Score. Main effects only; interaction terms were excluded due to empty cells arising from domain–query-type confounding (medical and finance domains contain only single-hop queries). η_p^2 = partial eta squared.

Effect	F	df _{num}	df _{den}	p	η_p^2
Chunking strategy	3.63	5	6336	.003	.003 (negligible)
Query type	30.73	3	6336	<.001	.014 (small)
Domain	397.66	2	6336	<.001	.112 (medium)

Table 5: Tukey HSD post-hoc pairwise comparisons by domain for LLM-as-Judge Score. Mean difference = Group A – Group B (positive values indicate Group A scores higher). 95% CI = confidence interval for mean difference. Cohen’s d computed using pooled SD. ns = not significant.

Group A	Group B	Mean Diff	95% CI	d	p_{adj}
General	Finance	+0.257	[+0.234, +0.281]	0.689	<.001
Medical	Finance	+0.271	[+0.245, +0.298]	0.776	<.001
General	Medical	−0.014	[−0.037, 0.009]	0.053	.334

Table 6: Tukey HSD post-hoc pairwise comparisons by query type for LLM-as-Judge Score. Mean difference = Group A – Group B (positive values indicate Group A scores higher). 95% CI = confidence interval for mean difference. Cohen’s d computed using pooled SD. ns = not significant.

Group A	Group B	Mean Diff	95% CI	d	p_{adj}
Comparative	Multi-hop	+0.071	[+0.021, 0.121]	0.251	.002
Comparative	Single-hop	+0.148	[+0.103, 0.193]	0.471	<.001
Comparative	Unanswerable	+0.126	[+0.074, 0.178]	0.436	<.001
Multi-hop	Single-hop	+0.077	[+0.047, 0.107]	0.228	<.001
Multi-hop	Unanswerable	+0.055	[+0.015, 0.094]	0.175	.002
Single-hop	Unanswerable	+0.022	[−0.011, 0.055]	0.065	.307

Table 7: Tukey HSD post-hoc pairwise comparisons by chunking strategy for LLM-as-Judge Score. Mean difference = Group A – Group B (positive values indicate Group A scores higher). 95% CI = confidence interval for mean difference. Cohen’s d computed using pooled SD. All significant pairs involve fixed_128 as the lower-scoring group; effect sizes are negligible throughout ($d < 0.20$), consistent with the small ANOVA $\eta_p^2 = .003$.

Group A	Group B	Mean Diff	95% CI	d	p_{adj}
fixed_256	fixed_128	+0.042	[+0.000, +0.084]	0.122	.049
fixed_512	fixed_128	+0.042	[+0.000, +0.084]	0.122	.049
semantic_128	fixed_128	+0.043	[+0.001, +0.085]	0.124	.046
semantic_256	fixed_128	+0.056	[+0.013, +0.099]	0.163	.003
semantic_512	fixed_128	+0.054	[+0.011, +0.097]	0.158	.004
fixed_512	fixed_256	+0.000	[−0.042, +0.042]	0.000	1.000
semantic_128	fixed_256	+0.001	[−0.041, +0.043]	0.003	1.000
semantic_256	fixed_256	+0.014	[−0.029, +0.057]	0.041	.936
semantic_512	fixed_256	+0.012	[−0.030, +0.055]	0.036	.963
semantic_128	fixed_512	+0.001	[−0.041, +0.043]	0.003	1.000
semantic_256	fixed_512	+0.014	[−0.029, +0.057]	0.041	.936
semantic_512	fixed_512	+0.012	[−0.030, +0.055]	0.036	.963
semantic_256	semantic_128	+0.013	[−0.030, +0.057]	0.038	.954
semantic_512	semantic_128	+0.012	[−0.032, +0.055]	0.033	.975
semantic_512	semantic_256	−0.002	[−0.046, +0.042]	0.005	1.000

Table 8: Welch’s t -test: Finance vs. General on (LLM Judge Score – Faithfulness). Negative mean difference indicates Finance scores are lower on this combined metric. Cohen’s d computed using pooled SD.

	Finance M	Finance SD	General M	General SD	t	p
Judge – Faithful.	−0.120	0.524	+0.025	0.365	−9.90	<.001

$N_{\text{Finance}} = 1600$, $N_{\text{General}} = 2992$, Welch $df = 2445.9$, Cohen’s $d = -0.322$ (small).